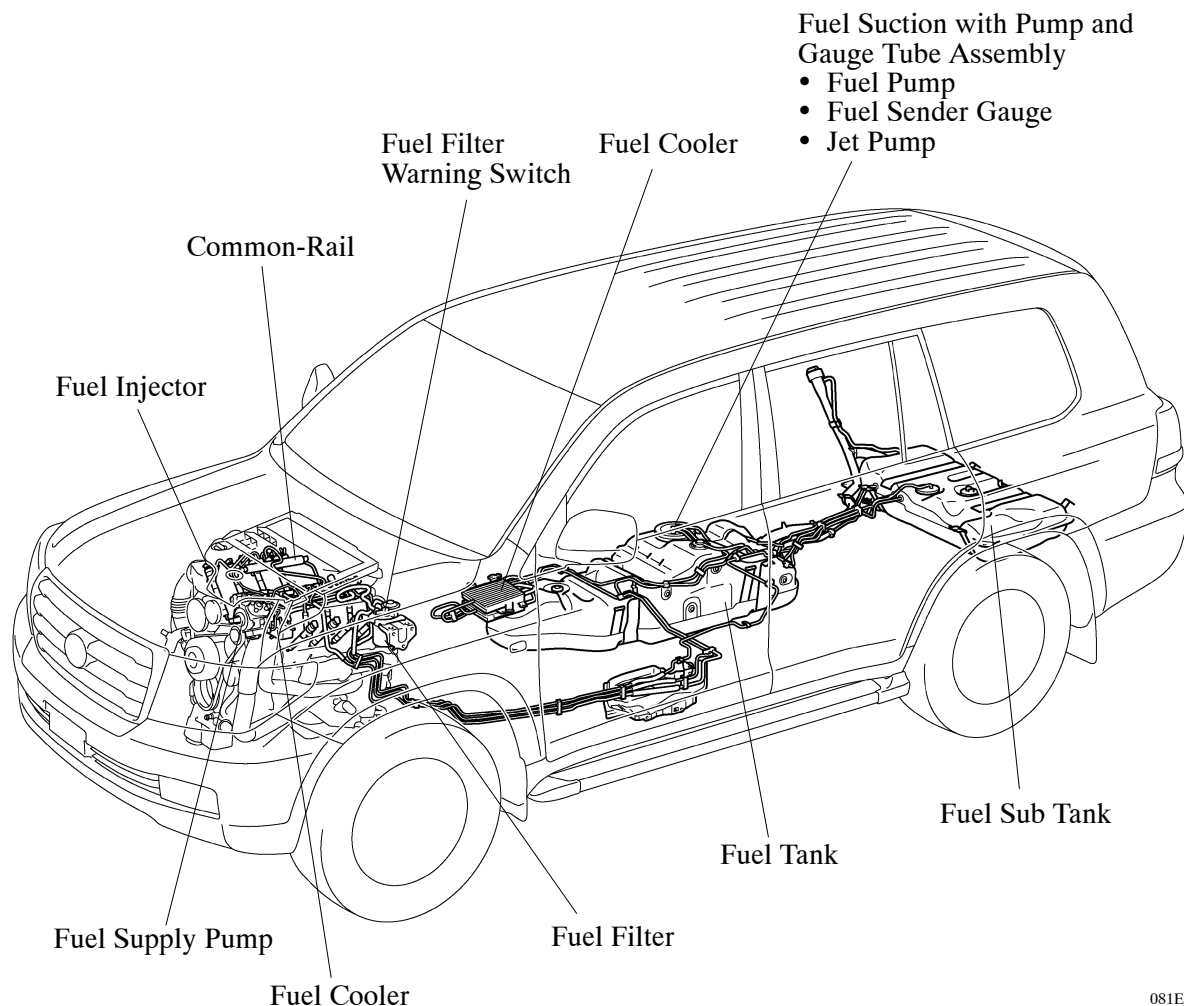


■ FUEL SYSTEM

1. General

- A common-rail system is used in the fuel injection system.
- 3-plunger type (HP4 type) fuel supply pump, which is compact and lightweight, is used.
- An injector compensation value and a QR code are printed on the fuel injector to realize high-precision control.
- A new type of fuel filter, in which the fuel filter element alone can be replaced, is used.
- A fuel filter warning switch is provided in the fuel filter.
- A recirculation type fuel heater is provided in the fuel filter as standard on European models and as an option on general country models.
- 2 fuel coolers are used in the fuel return path.
- Quick connectors are used to connect the fuel lines for ease of serviceability.
- The dual fuel tank comprising a fuel tank and fuel sub tank is provided as standard on Australian* and G.C.C. country models and as an option on general country models. A multi-layer plastic fuel tank is used for the fuel tank. (The fuel sub tank however is made of steel.)
- A fuel pump and jet pump are provided in the fuel tank on models equipped with a dual fuel tank to transfer fuel from the fuel sub tank to the fuel tank.

*: On VX grade models, a single fuel tank is available as an option.



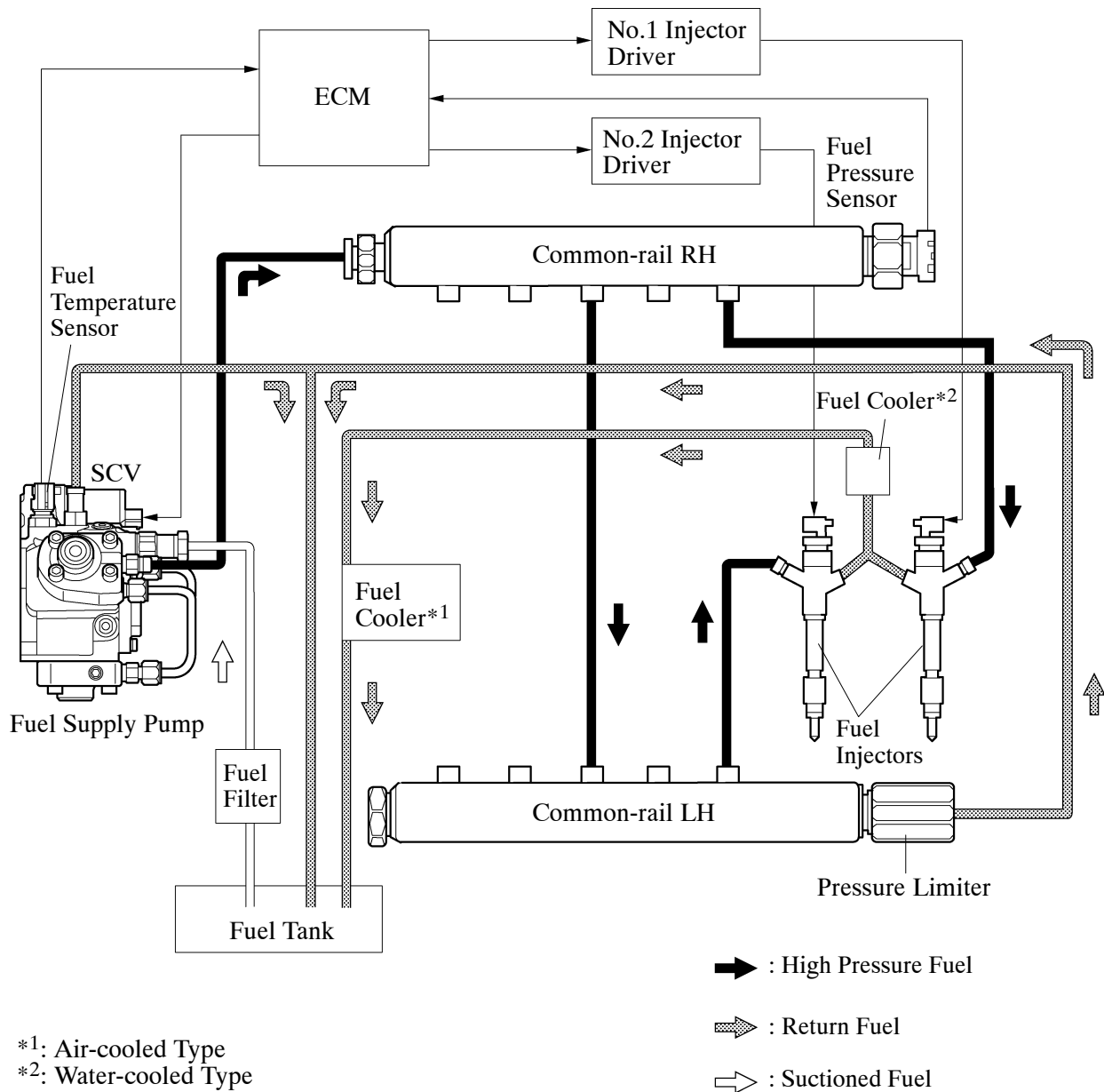
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2. Common-rail System

General

In this system, the high-pressurized fuel that is supplied by the fuel supply pump is stored in the common-rail, and the ECM sends signals to the injectors via the injector driver in order to control the injection timing and injection volume. For details of this control, see page EG-192.

► System Diagram ◀



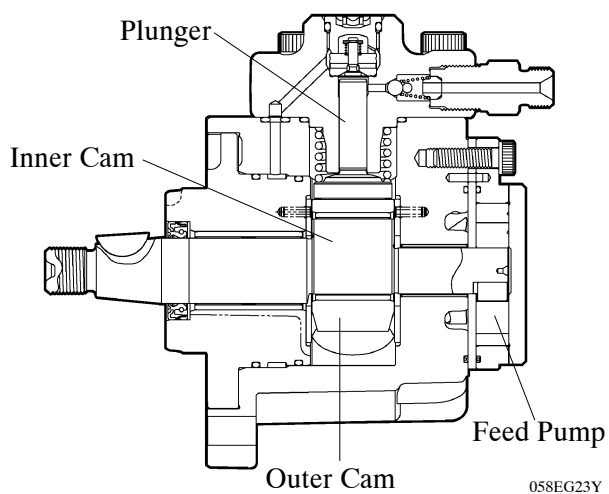
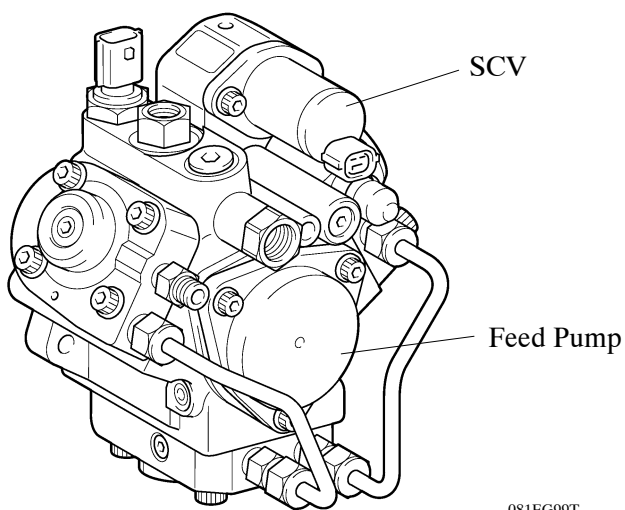
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Models without Fuel Heater

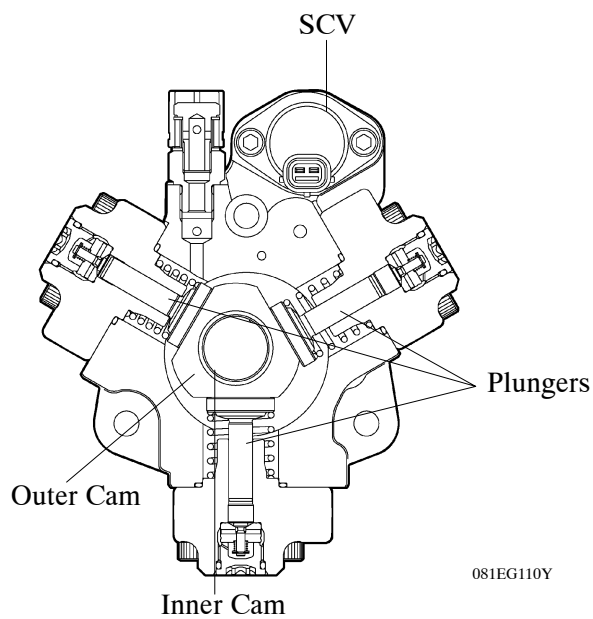
Fuel Supply Pump

1) General

The fuel supply pump consists of an inner cam (eccentric cam), outer cam (polygon ring), and 3 plungers, SCV (Suction Control Valve), and a feed pump. Each plunger is placed outside of the outer cam.



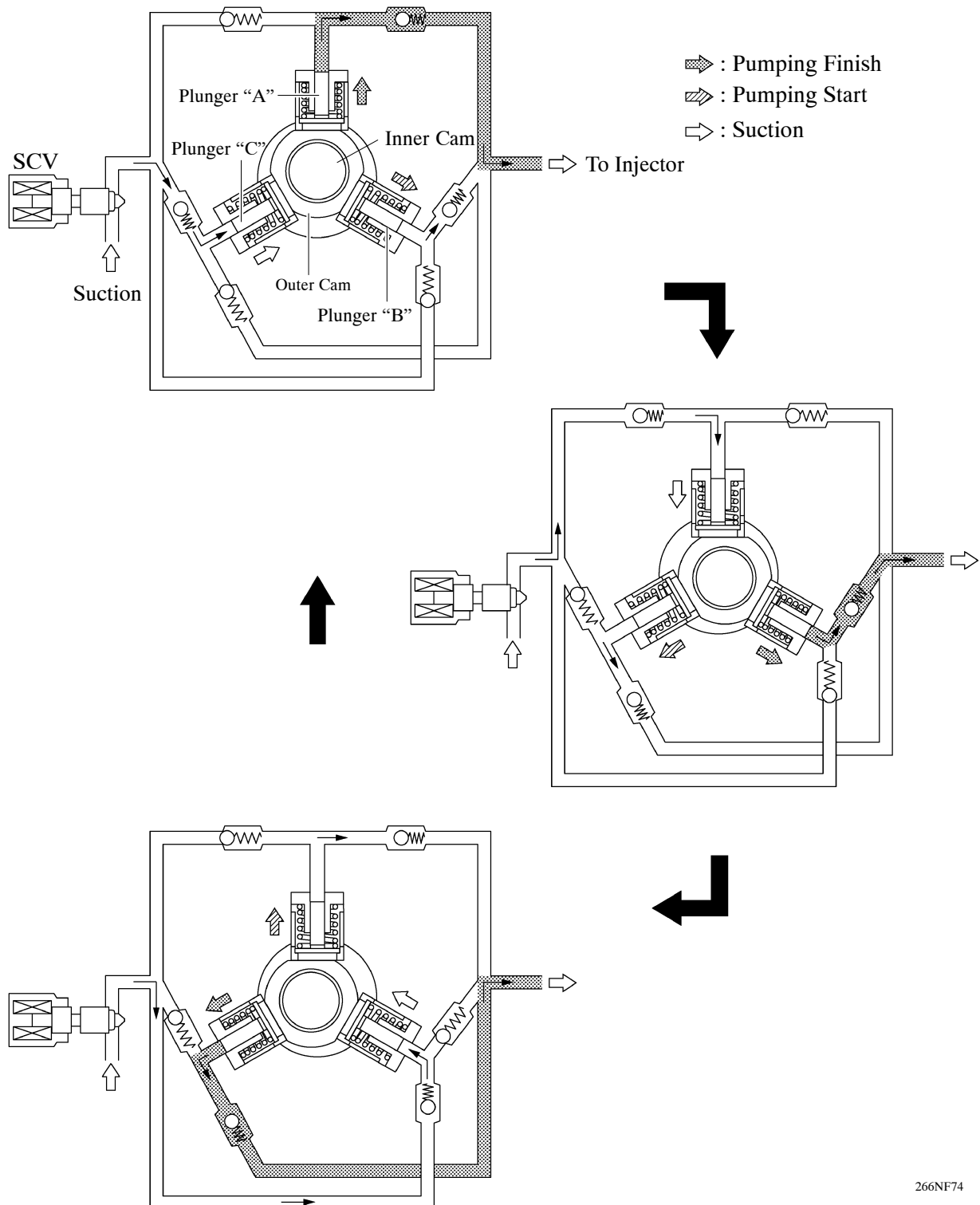
Cross Section



Cross Section

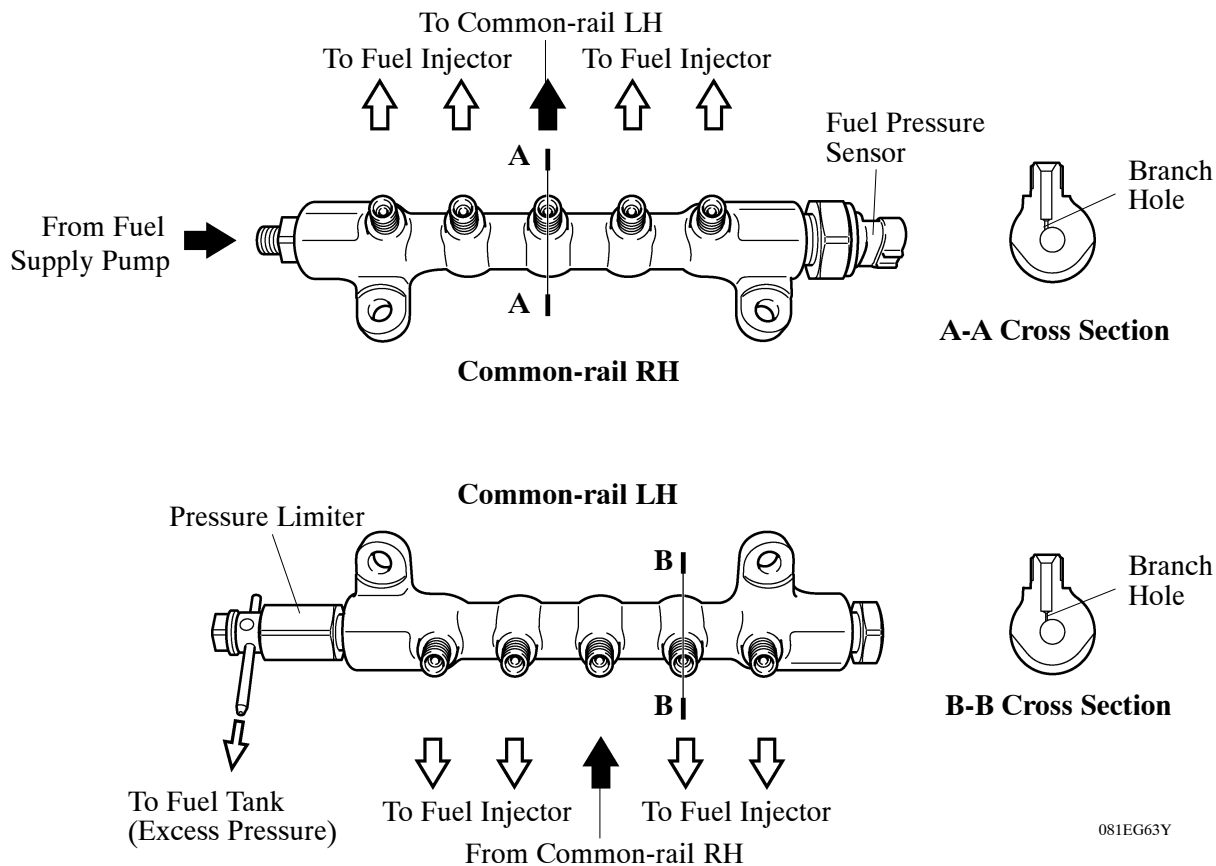
2) Operation

- Due to the rotation of the inner cam (eccentric cam), the outer cam pushes plunger “A” upward as illustrated below. The force of the spring pulls plungers “B” and “C”. As a result, plungers “B” and “C” draw fuel in, and plunger “A” pumps fuel at the same time.
- The SCV controls the volume of fuel that is drawn into the plungers in accordance with the signals from the ECM.



Common-rail

- The function of the common-rail is to store the fuel that has been pressurized by the fuel supply pump. The common-rail is provided with a fuel pressure sensor, which detects the fuel pressure in the common-rail, and a pressure limiter that mechanically relieves the pressure in case the internal pressure of the common-rail rises abnormally.
- Internally, the common-rail contains a main hole and five branch holes that intersect the main hole. Each branch hole functions as an orifice that dampens the fluctuation of the fuel pressure.



Service Tip

- Fuel pressure sensor has its sealing portion plastic-deformed in order to keep sealing performance, so do not reuse it after disassembling.
- The fuel pressure sensor and pressure limiter are supplied as a set with the common-rail.
- Do not disassemble the pressure limiter.
- If parts that affect the alignment have been changed, make sure to replace the pipe with a new one as well. The parts that require the replacement of a pipe are listed below.

Injection Pipe: Fuel Injector, Common-rail, Cylinder Head and Intake Manifold

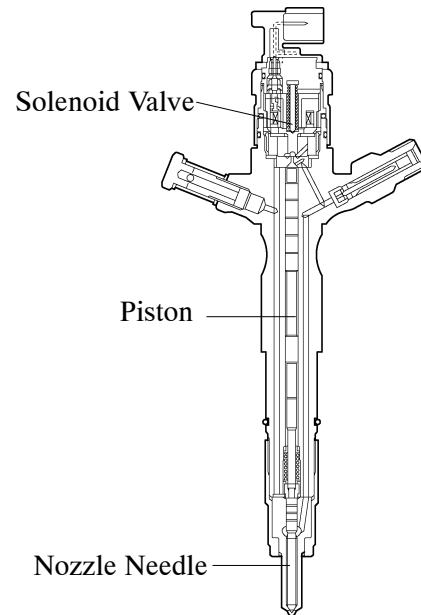
Fuel Inlet Pipe: Fuel Supply Pump, Common-rail, Cylinder Head and Intake Manifold

For details, refer to the Land Cruiser (Station Wagon) Repair Manual (Pub. No. RM0810E).

Fuel Injector

1) General

- The fuel injector consists of a nozzle needle, piston, and solenoid valve.
- An injector compensation value and QR (Quick Response) code containing encoded characteristics of the fuel injector are printed on each fuel injector.
- The injector compensation value and QR code contain various pieces of information regarding the fuel injector, such as model code, injection volume correction, and injection timing correction values.

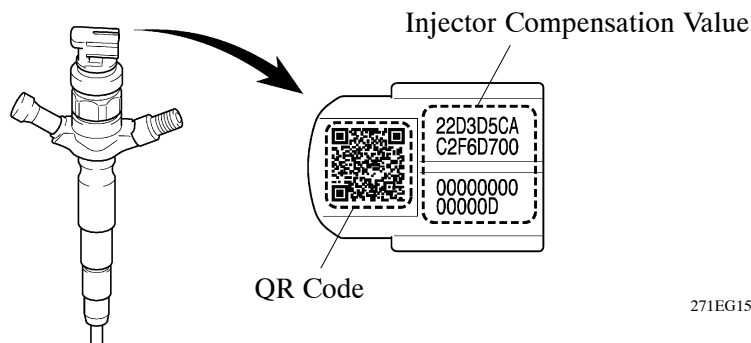


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Service Tip

- If the ECM is replaced, use the intelligent tester II and input the injector compensation values of all 8 injectors. If one of the fuel injectors is replaced, input the injector compensation value of the replaced fuel injector. Then, the proper compensation will be made so that the injection volume precision prior to the replacement will remain unchanged. For details, refer to the Land Cruiser (Station Wagon) Repair Manual (Pub. No. RM0810E).
- The QR code, which requires a special scan tool, is not used at TOYOTA dealers.



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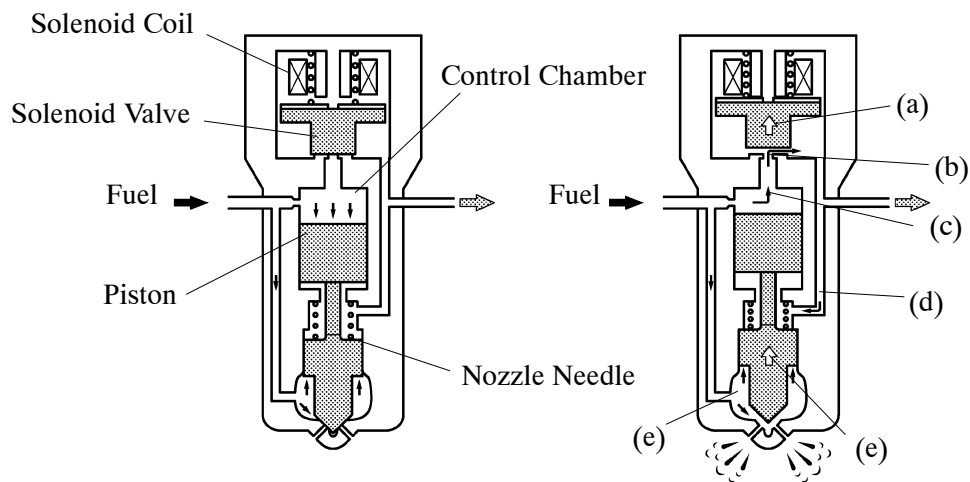
—REFERENCE—

What is QR (Quick Response) Code?

- QR code, a matrix symbology consisting of an array of nominally square cells, allows omni-directional, high-speed reading of large amounts of data.
- QR code encodes many types of data such as numeric, alphanumeric, kanji, kana and binary code. A maximum of 7,089 characters (numeric) can be encoded.
- QR code (2D code) contains information in the vertical and horizontal direction, whereas a bar code contains data in one direction only. QR code (2D code) holds a considerably greater volume of information than a bar code.

2) Operation

- (a) When electrical current is applied to the solenoid coil, it pulls the solenoid valve up.
- (b) The orifice of the control chamber opens, allowing the fuel to flow out.
- (c) The fuel pressure in the control chamber drops.
- (d) Simultaneously, fuel flows from the orifice to the bottom of the piston and raises the piston up (to enhance response).
- (e) As a result, the piston raises the nozzle needle to inject fuel.



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3. Fuel Tank

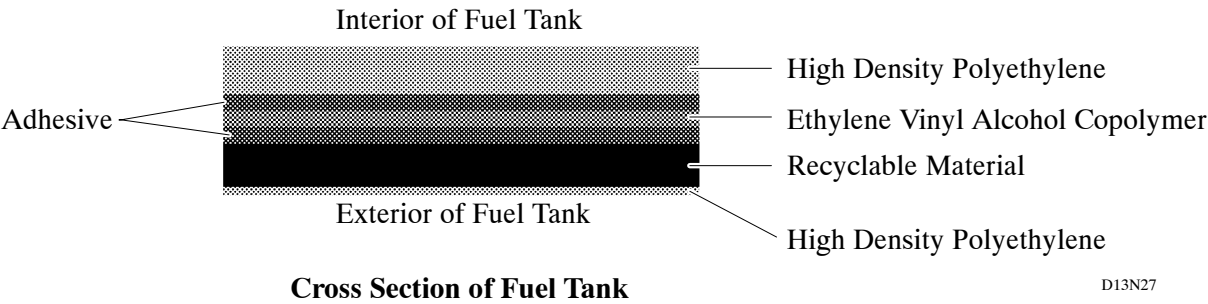
General

- A jet pump, which automatically transfers fuel from the fuel sub tank to the fuel tank, and a fuel pump, which drives the jet pump, are provided in the fuel tank on models equipped with a dual fuel tank.
- The multi-layer plastic fuel tank consists of six layers of four types of materials, and one of those is a recyclable material to address environmental concerns.

► Fuel Tank Capacity ◀

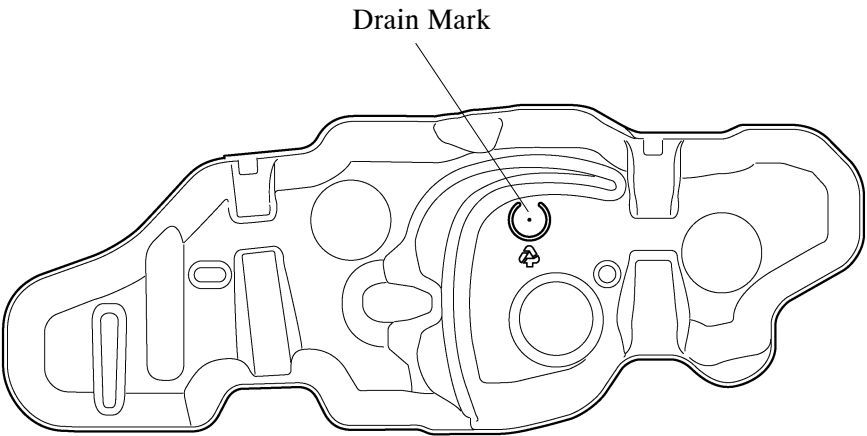
	Liters (US qts, Imp qts)
Models with Dual Fuel Tank	138 (145.8, 121.4)
Models with Single Fuel Tank	93 (98.3, 81.8)

EG



Service Tip

Drain marks have been provided at the lowest position of the fuel tank. When dismantling (scraping) the vehicle, drain fuel by drilling holes at the drain mark.

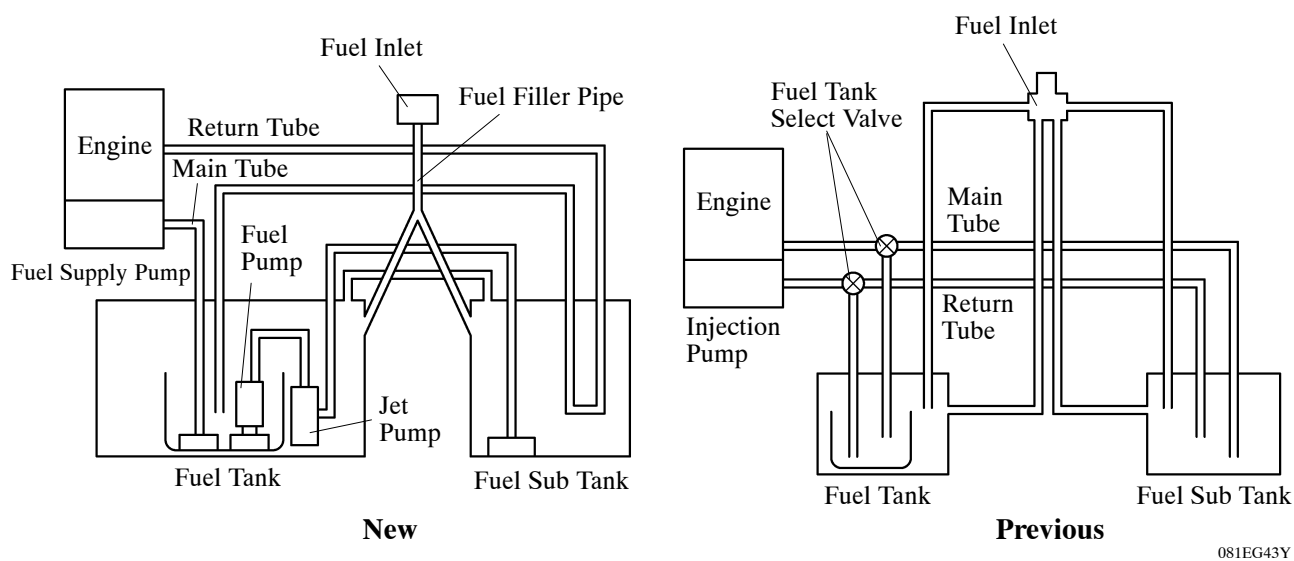


Construction of Dual Fuel Tank

1) General

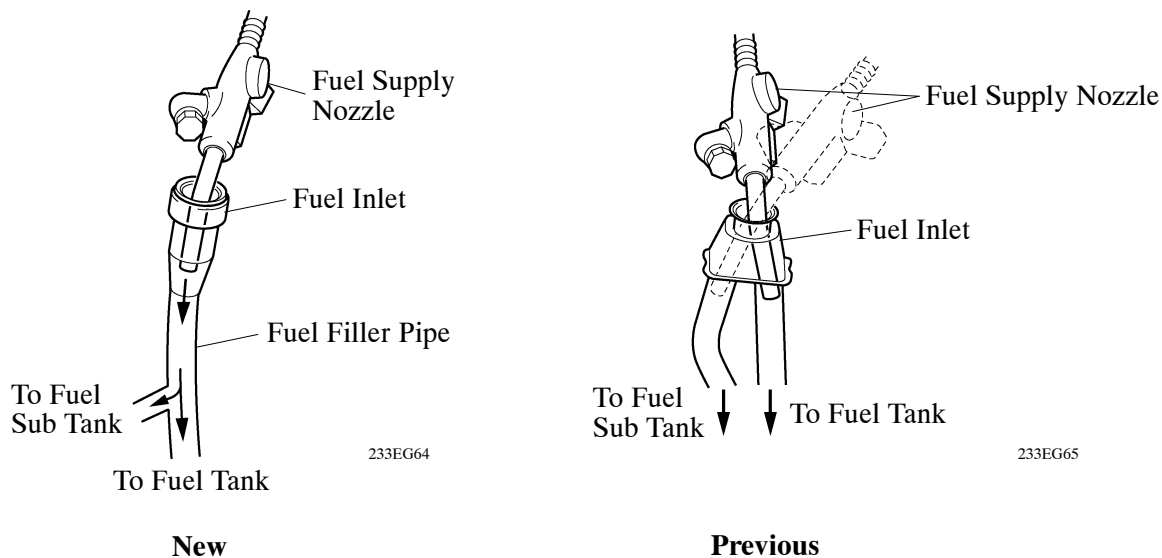
- On the previous model, either the fuel tank or fuel sub tank, from which fuel is supplied, is selected by operating the sub fuel tank switch. On the new model however, a jet pump and fuel pump are provided in the fuel tank in order to automatically transfer fuel from the fuel sub tank to the fuel tank.
- Fuel pump operation is controlled by the ECM in accordance with the actual fuel amount in the fuel tanks. For details on fuel pump control, see page EG-205.
- For the fuel filler pipe, the structure that is united right under the fuel inlet and forked into two in the middle is used in its configuration. Therefore, the driver can refuel without the need to distinguish the fuel tank from the fuel sub tank.

► System Diagram ◀



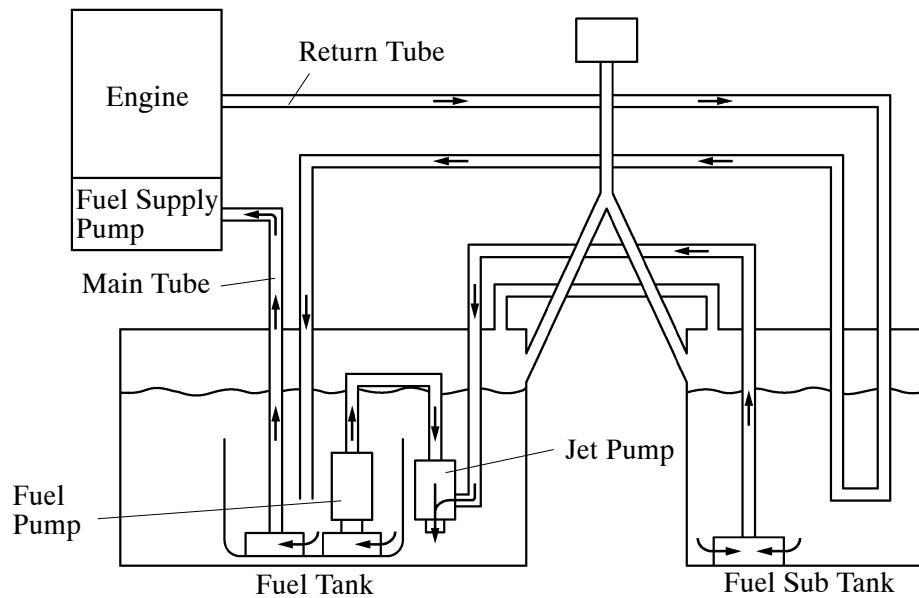
2) Fuel Inlet

- On the previous model, the fuel filler pipe has branched into two pipes directly below the fuel inlet. Therefore, the driver had to select whether to fill the fuel tank or the fuel sub tank during refueling.
- On the new model, there is only one fuel filler pipe directly below the fuel inlet, which branches into two pipes at a midway point.



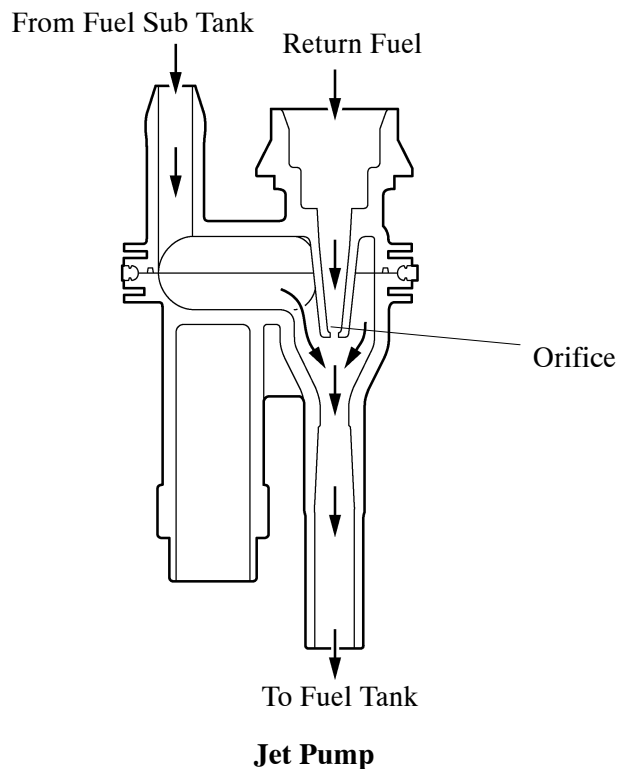
3) Operation

- The fuel from the fuel pump actuates the jet pump in the fuel tank, in order to draw fuel from the fuel sub tank to the fuel tank.



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- The fuel from the fuel pump passes through the orifice in the jet pump and returns to the fuel tank. Because the flow speed of the fuel from the fuel pump increases as it passes through the orifice, a vacuum is created near the exit of the orifice. This vacuum causes the fuel to be drawn from the fuel sub tank to the fuel tank.

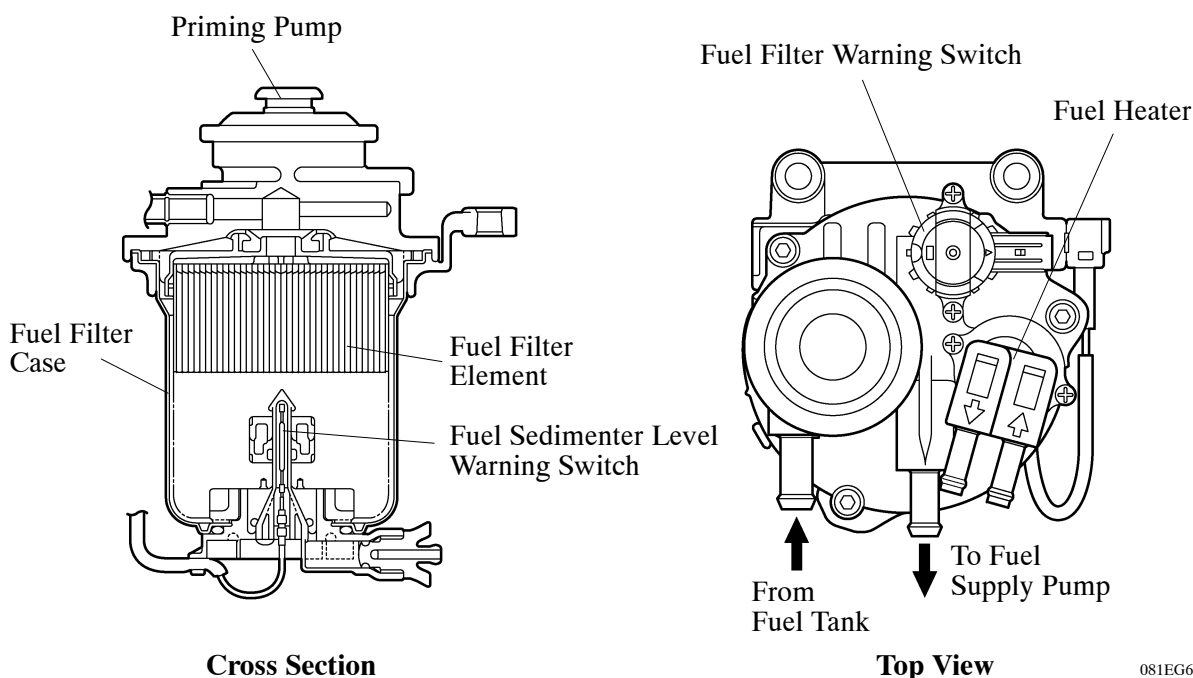


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4. Fuel Filter

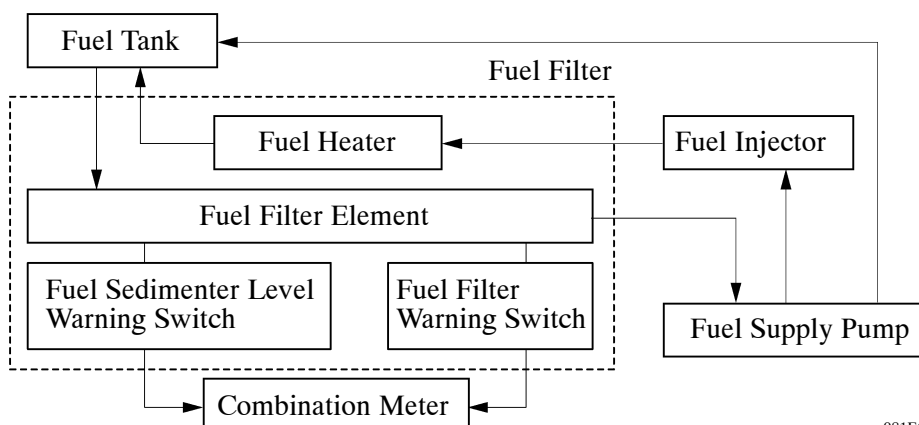
General

- While the fuel filter element of the conventional fuel filter has been integrated with the fuel filter case, the new fuel filter consists of a separate fuel filter element and fuel filter case. A paper filter element that offers high filtering efficiency and captures the minutest particles is used.
- A fuel filter warning switch, which turns ON/OFF when the internal vacuum of the filter increases, is provided in the fuel filter.
- Instead of the electrical type fuel heater used on the conventional model, a recirculation type fuel heater is provided as standard on European models and as an option on general country models, in order to improve the fuel heating performance.]



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► System Diagram ◀



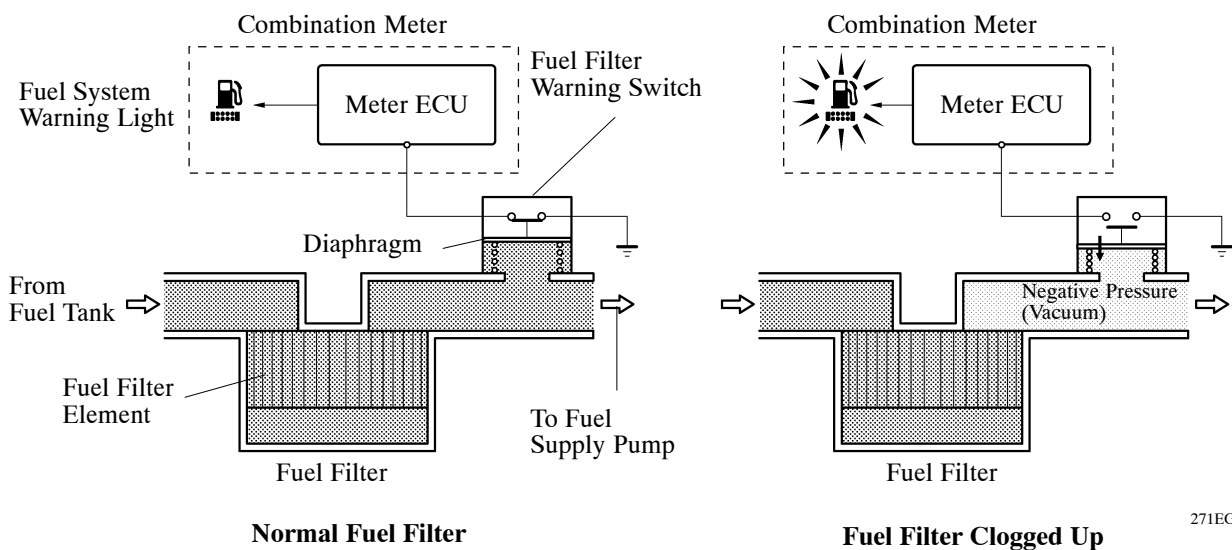
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Fuel Filter Warning Switch

- A fuel filter warning switch, which turns ON/OFF when the internal vacuum of the filter increases, is provided in the fuel filter. This switch, which turns OFF when the internal vacuum of the fuel filter increases to a predetermined level, is connected by wire to the meter ECU.
- When the meter ECU detects that the internal vacuum of the fuel filter has increased (by way of the fuel filter warning switch OFF signal), it determines that the fuel filter has become clogged. Then it illuminates the fuel system warning light*¹ (or displays a message on the multi-information display*²) on the combination meter to urge the driver to replace the fuel filter. For details on fuel filter warning light control, see page BE-48.

*1: Only on models with analog type combination meter

*2: Only on models with optitron display type combination meter



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Fuel Sedimenter Level Warning Switch

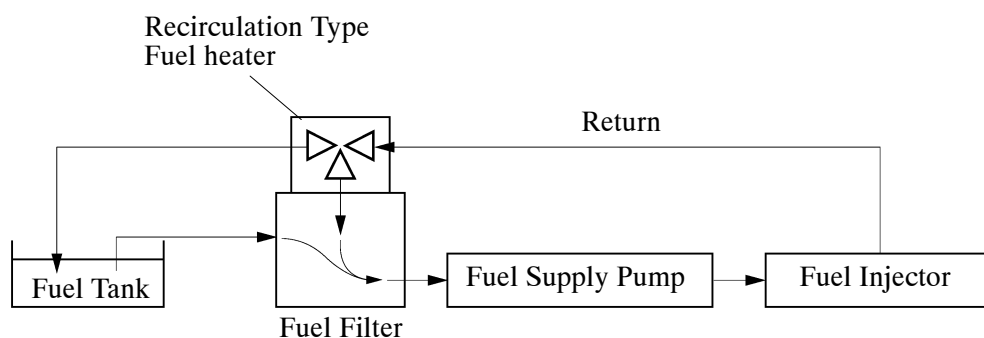
- When the water in the sedimenter section reaches a certain amount, the level warning switch comes ON and the fuel system warning light in the combination meter blinks*¹ (or a message is displayed on the multi-information display*²).
- On models except for Europe, when the fuel system warning light blinks, the buzzer in the combination meter sounds.
- For details on fuel filter warning light control, see page BE-48.

*1: Only on models with analog type combination meter

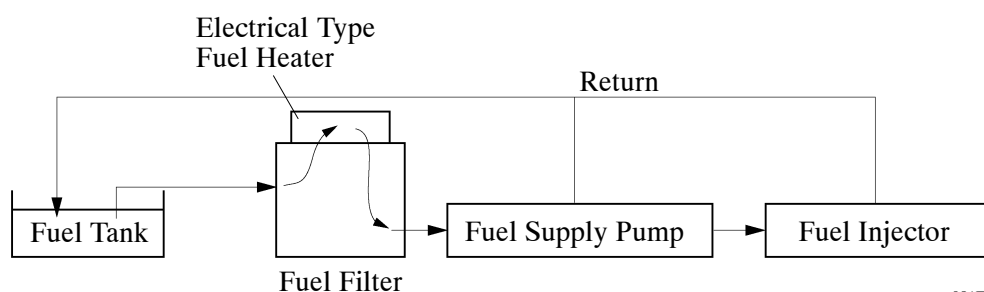
*2: Only on models with optitron display type combination meter

Fuel Heater

- This heater heats the fuel by recirculating the returned fuel, which becomes to have a high temperature by the fuel pumping of the fuel supply pump, to the fuel filter.
- The fuel heater uses a recirculation valve (bimetal), which is mounted on the fuel return line and opens and closes in accordance with the fuel temperatures to draw return fuel to the fuel filter.
- While the engine is cold, the recirculation valve inside the fuel heater will open due to the bimetal action, so the return fuel will return to the fuel filter and the fuel tank.
- While the engine is warm, the recirculation valve will close due to the bimetal action, so all the return fuel will return to the fuel tank.



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New

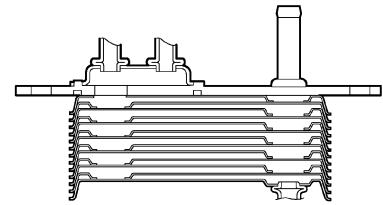
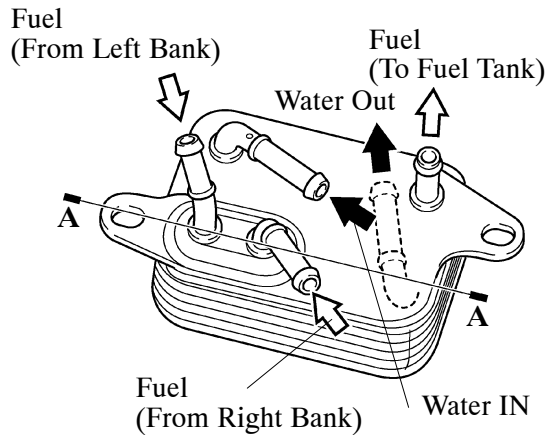
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Conventional

5. Fuel Cooler

The two types of the fuel coolers are used; one is the water-cooled type, and the other is the air-cooled type. These fuel coolers, which are provided in the fuel return path, cool the return fuel that has reached a high temperature as a result of the pumping of the fuel supply pump. This prevents the reduction in fuel viscosity that is caused by the rise in fuel temperature, and improves the reliability of the fuel system.

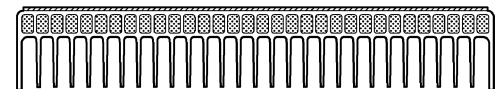
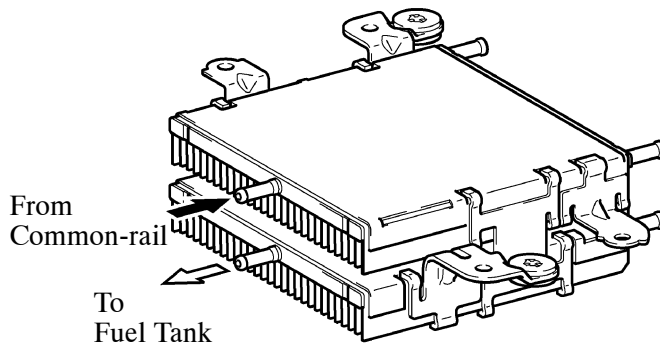
► Water-cooled Type Fuel Cooler ◀



A - A Cross Section

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► Air-cooled Type Fuel Cooler ◀



Fuel Cooler Cross Section

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